2.4 Sum and Difference Formulas

Name _____ Period ____ Date _____

Pre-Calculus 2.4 (30-minute plan) · Student sheet

I can…

- use sum and difference formulas to find exact values.
- use sum and difference formulas to rewrite and evaluate expressions.
- use sum and difference formulas to solve equations.

Vocabulary · fill in the missing word

Sum and difference formulas — $\sin(u \pm v) =$ _____; $\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$; $\tan(u \pm v) = (\tan u \pm \tan v)/(1 \mp \tan u \tan v)$

Part 1 · Notes and examples (fill in each note, then work the example)

Sine formulas: $\sin(u + v) =$ _____ $\sin(u - v) =$ _____

Example 1

Find the exact value of $\sin 75°$.

Cosine formulas: $\cos(u + v) =$ _____ $\cos(u - v) =$ _____

Tangent formulas: $\tan(u + v) =$ _____ $\tan(u - v) =$ _____

The signs: In the sine formulas the sign _____; in the cosine formulas the sign _____.

Example 2

Write $\sin 42° \cos 12° - \cos 42° \sin 12°$ as a trigonometric function of one angle. Then find its exact value.

Example 3

$\sin u = \frac{5}{13}$ with $0 < u < \frac{\pi}{2}$, and $\cos v = -\frac{3}{5}$ with $\frac{\pi}{2} < v < \pi$. Find the exact value of $\sin(u + v)$.

Part 2 · Partner practice

Partner practice

Find the exact value of cos 105°.